Semantic Clustering of Agricultural Imagery Using Vision-Language Embeddings: Drone and Ground Camera Multi-Site Analysis

Research Notes - Draft v0.8

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Abstract

This study explores the application of vision-language embedding models for automatic semantic categorization of agricultural imagery across multiple field sites and capture modalities. Using CLIP (Contrastive Language-Image Pre-training) embeddings with UMAP dimensionality reduction, we demonstrate that both **drone** and **ground camera** images can be automatically clustered by semantic content without manual labeling or domain-specific training.

Drone Imagery

Analysis of two drone survey datasets from Chilean tomato fields reveals consistent clustering behavior:

- **Santa Ines Drone** (97 images): Two clusters identified—close-up crop inspection (92%) and high-altitude field mapping (8%)—with 3.56 UMAP units separation.
- **La Capilla Drone** (111 images): Two clusters identified—very close-up plant-level (23%) and medium-altitude row-visible (77%)—with 12.93 UMAP units separation and perfect cluster isolation (0% cross-category connections).

Ground Camera Imagery

Four ground-level datasets captured with mobile phones ("Celular") were also analyzed:

- **Santa Ines Ground** (635 images): Ground-level crop inspection imagery
- **La Capilla Ground** (822 images): Ground-level field documentation
- **Apalta** (961 images): Extensive ground survey with iPhone HEIC format
- **Camarico** (33 images): Compact ground-level dataset

Key Findings

Embedding-based clusters capture **visual semantic similarity** rather than geographic proximity—images from scattered GPS locations cluster together based on content type. The methodology generalizes across both aerial (drone) and terrestrial (ground camera) capture modalities, suggesting that pretrained vision-language models offer a practical, zero-configuration approach for organizing diverse agricultural image datasets.

Contents

1 Introduction 4
- 1.1 Background . . . 4
- 1.2 Vision-Language Models for Image Understanding . . . 4
- 1.3 Research Gap . . . 4

2 Objectives 4
2.1 General Objective . . . 4
2.2 Specific Objectives . . . 5
2.3 Hypotheses . . . 5

3 Materials and Methods 5
3.1 Study Areas and Data Collection . . . 5
3.1.1 Field Sites . . . 5
3.1.2 Capture Modalities . . . 5
3.1.3 Drone Imagery Datasets . . . 6
3.1.4 Ground Camera Imagery Datasets . . . 6
3.2 Software Environment . . . 6
3.2.1 Dataset Management . . . 6
3.2.2 Analysis Tools . . . 7
3.3 Embedding Computation . . . 7
3.3.1 Model Selection . . . 7
3.3.2 Dimensionality Reduction . . . 7
3.3.3 Implementation . . . 7
3.4 Clustering and Classification . . . 8
3.5 GPS Extraction . . . 8
3.6 Geospatial Data Export . . . 8
3.7 Similarity Network Analysis . . . 8
3.8 Statistical Analysis . . . 8

4 Results 9
4.1 Part I: Drone Imagery Analysis . . . 9
4.2 Drone Site 1: Santa Ines . . . 9
4.2.1 Combined Spatial and Visual Analysis . . . 9
4.2.2 Cluster Characteristics . . . 10
4.2.3 Similarity Network . . . 10
4.3 Drone Site 2: La Capilla . . . 10
4.3.1 Combined Spatial and Visual Analysis . . . 10
4.3.2 Visual Characteristics . . . 11
4.3.3 Cluster Characteristics . . . 12
4.3.4 Similarity Network . . . 13
4.3.5 GPS vs Embedding Space . . . 13
4.4 Cross-Site Comparison . . . 14
4.4.1 Summary Statistics . . . 14
4.4.2 Key Findings . . . 14
4.4.3 Cluster Cohesion Comparison . . . 15
4.5 Part II: Ground Camera Imagery Analysis . . . 15
4.5.1 Dataset Summary . . . 15
4.5.2 Embedding Space Analysis . . . 15
4.5.3 Ground Site 1: Santa Ines (Celular) . . . 16
4.5.4 Ground Site 2: La Capilla (Celular) . . . 17
4.5.5 Ground Site 3: Apalta . . . 18
4.5.6 Ground Site 4: Camarico . . . 19
4.5.7 Ground Site 5: Lo Castillo . . . 20
4.5.8 Lo Castillo Cluster Characteristics . . . 21
4.5.9 Multi-Format Analysis . . . 21
4.5.10 Ground Camera: Network Analysis Summary . . . 22
4.5.11 Temporal Clustering Discovery . . . 22

4.5.12 Comparison: Drone vs Ground Camera . . . 22
4.5.13 Key Characteristics . . . 23
4.5.14 GPS Coverage . . . 23
4.5.15 Geospatial Outputs . . . 23
4.6 Part III: Combined Dataset Summary . . . 23
4.6.1 Data Availability by Site and Modality . . . 24
4.6.2 Drone vs Ground Camera: Side-by-Side Comparison . . . 24

5 Discussion **25**
5.1 Interpretation of Results . . . 25
5.1.1 Semantic Clustering Without Domain-Specific Training . . . 25
5.1.2 Embedding Space vs. Geographic Space . . . 25
5.1.3 Site-Specific Clustering Patterns . . . 25
5.1.4 Cluster Cohesion Asymmetry . . . 26
5.1.5 Ground Camera: Temporal Sensitivity Discovery . . . 26
5.1.6 Multi-Modal Clustering Patterns . . . 26
5.2 Practical Implications . . . 27
5.2.1 Automated Dataset Curation . . . 27
5.2.2 Content-Based Image Retrieval . . . 27
5.2.3 Multi-Site Workflows . . . 27
5.3 Data Availability Considerations . . . 27
5.3.1 Incomplete Multi-Modal Coverage . . . 27
5.3.2 Dataset Size Variability . . . 28
5.4 Limitations . . . 28
5.5 Future Directions . . . 28

6 Conclusions **29**

A Observation Log **30**

B Technical Details **31**
B.1 Software Versions . . . 31
B.2 Data Paths . . . 31
B.3 Key Commands . . . 32
B.3.1 FiftyOne Dataset Operations . . . 32
B.3.2 Compilation . . . 32
B.4 FiftyOne Brain Keys . . . 33
B.5 GPS Extraction from Drone Images . . . 33
B.6 Satellite Map Integration . . . 33
B.6.1 Coordinate Projection . . . 33
B.7 Analysis Scripts Reference . . . 34

1 Introduction

1.1 Background

Unmanned Aerial Vehicles (UAVs) have become essential tools in precision agriculture, enabling high-resolution monitoring of crop health, irrigation patterns, and field conditions. Modern drone surveys generate large volumes of imagery—often hundreds to thousands of images per flight session—creating significant data management challenges.

[TODO: Add citations: drone imagery in agriculture, data volume challenges**]**

A typical agricultural drone survey may include:

- **Systematic mapping flights**: High-altitude images for orthomosaic generation
- **Targeted inspection flights**: Low-altitude close-ups of specific crop areas
- **Oblique/transitional images**: Various angles during ascent/descent

Organizing this heterogeneous imagery typically requires manual review or relies solely on GPS metadata, which captures *where* images were taken but not *what* they contain semantically.

1.2 Vision-Language Models for Image Understanding

Recent advances in vision-language models, particularly CLIP (Contrastive Language-Image Pre-training), have demonstrated remarkable zero-shot capabilities for image understanding. These models learn joint embeddings of images and text, enabling semantic similarity comparisons without task-specific training.

[OBS: CLIP embeddings capture semantic content (altitude/purpose) rather than just spatial position—this is a key distinction from GPS-based organization**]**

1.3 Research Gap

While vision-language models have been extensively studied for general image classification and retrieval, their application to agricultural drone imagery organization remains underexplored. Specifically:

1. Can pretrained models (without agricultural fine-tuning) meaningfully cluster drone imagery?
2. What semantic distinctions do these models capture in agricultural contexts?
3. How do embedding-based clusters compare to traditional metadata-based organization?

[TODO: Expand literature review: existing work on drone image organization, CLIP applications in remote sensing**]**

2 Objectives

2.1 General Objective

To evaluate the effectiveness of vision-language embedding models for automatic semantic organization of agricultural drone imagery, with application to precision agriculture data management workflows.

2.2 Specific Objectives

1. **Embedding Analysis**: Compute and visualize CLIP embeddings for drone survey imagery to identify natural semantic clusters.
2. **Cluster Characterization**: Analyze the semantic distinctions captured by embedding-based clusters (e.g., altitude, crop visibility, image purpose).
3. **Comparison with Metadata**: Compare embedding-based organization with GPS/EXIF metadata-based approaches to understand complementary information.
4. **Similarity Search**: Implement and evaluate content-based image retrieval for finding similar drone images within large datasets.
5. **Workflow Integration**: Develop practical tools for integrating embedding-based organization into agricultural data pipelines.

2.3 Hypotheses

H1 Pretrained CLIP embeddings will separate drone images by semantic content (flight altitude, imaging purpose) without domain-specific training.

H2 Embedding-based clusters will capture information orthogonal to GPS position—images from different locations but similar purposes will cluster together.

H3 Similarity search based on embeddings will retrieve semantically relevant images more effectively than spatial proximity alone.

[**TODO:** Refine hypotheses based on preliminary results]

3 Materials and Methods

3.1 Study Areas and Data Collection

3.1.1 Field Sites

Four agricultural field sites in central Chile were surveyed:

Table 1: Study site characteristics

Site	GPS Center	Crop	Survey Date
Santa Ines	-34.4746, -70.9516	Tomato	January 2026
La Capilla	-34.4722, -70.9337	Tomato	January 2026
Apalta	-34.4743, -70.9531	Tomato	January 2026
Camarico	-34.3580, -70.8918	Tomato	January 2026

3.1.2 Capture Modalities

Two distinct capture modalities were employed to document field conditions:

1. **Drone (Aerial)**: DJI drones capturing overhead imagery at varying altitudes
2. **Ground Camera (Celular)**: Mobile phones capturing ground-level imagery while walking through fields

[**OBS:** The combination of drone and ground camera imagery provides complementary perspectives—aerial for field-scale patterns, ground-level for plant-specific detail]

3.1.3 Drone Imagery Datasets

Table 2: Drone imagery dataset characteristics

Site	Images	Format	Resolution	Total Size
Santa Ines	97	JPEG	5472×3648	~1.1 GB
La Capilla	111	JPEG	5472×3648	~1.4 GB

Drone imagery characteristics:

- **Platform**: DJI drone with integrated GPS
- **File format**: JPEG, approximately 10-13 MB per image
- **Embedded metadata**: GPS coordinates, timestamp, altitude, camera settings
- **Perspective**: Top-down aerial view at varying altitudes (close-up to overview)

3.1.4 Ground Camera Imagery Datasets

Table 3: Ground camera imagery dataset characteristics

Site	Images	Format	Device	GPS Coverage
Santa Ines	635	JPEG	Android	100% (635/635)
La Capilla	822	JPEG	Android	97% (799/822)
Apalta	961	HEIC	iPhone XR	99.9% (960/961)
Camarico	33	JPEG	Android	82% (27/33)

Ground camera imagery characteristics:

- **Platform**: Mobile phones (Android and iPhone)
- **File format**: JPEG (Android) or HEIC (iPhone)
- **Typical size**: 5-7 MB per image (JPEG), 2-3 MB per image (HEIC)
- **Embedded metadata**: GPS coordinates, timestamp, device info
- **Perspective**: Ground-level, horizontal view of crop rows and individual plants

[FIND: Ground camera datasets are significantly larger (635-961 images) than drone datasets (97-111 images), reflecting the different data collection workflows—walking through fields vs. programmed flight paths**]**

3.2 Software Environment

3.2.1 Dataset Management

FiftyOne v1.12.0 was used for dataset management, embedding computation, and visualization. FiftyOne provides:

- Efficient handling of large image datasets (2000+ images across all sites)
- Built-in integration with embedding models
- Interactive visualization tools
- Similarity search functionality

3.2.2 Analysis Tools

- **Python 3.12**: Data processing and embedding computation
- **R 4.3.3**: Statistical analysis and visualization (ggplot2)
- **PyTorch**: Deep learning backend for CLIP model
- **PIL/Pillow**: EXIF metadata extraction (JPEG)
- **pillow-heif**: HEIC format support for iPhone images

3.3 Embedding Computation

3.3.1 Model Selection

CLIP ViT-B/32 (Vision Transformer, Base, 32×32 patches) was selected for embedding computation. This model:

- Provides 512-dimensional embeddings per image
- Was pretrained on 400M image-text pairs
- Offers good balance of computational efficiency and semantic richness
- Requires no domain-specific fine-tuning

3.3.2 Dimensionality Reduction

UMAP (Uniform Manifold Approximation and Projection) was applied to reduce embeddings from 512D to 2D for visualization:

$$\mathbf{z}_i = \mathrm{UMAP}(\mathbf{e}_i), \quad \mathbf{e}_i \in \mathbb{R}^{512}, \quad \mathbf{z}_i \in \mathbb{R}^2 \tag{1}$$

UMAP was run with default parameters (n_neighbors=15, min_dist=0.1, metric=euclidean).

3.3.3 Implementation

The same pipeline was applied to all datasets (both drone and ground camera):

Listing 1: Embedding computation pipeline

```
import fiftyone as fo
import fiftyone.brain as fob

# Create dataset from images
dataset = fo.Dataset.from_images_dir(path, name="site_name")

# Compute CLIP embeddings with UMAP projection
fob.compute_visualization(
    dataset,
    brain_key="clip_viz",
    model="clip-vit-base32-torch",
    method="umap"
)

# Compute similarity index for retrieval
fob.compute_similarity(
    dataset,
    brain_key="clip_similarity",
```

```
    embeddings="clip_viz"
)
```

3.4 Clustering and Classification

Image clusters were identified through visual inspection of the UMAP projection at each site. Clusters were characterized by:

1. **Visual inspection**: Examining representative images from each region
2. **Centroid analysis**: Computing cluster centroids and inter-cluster distances
3. **Manual tagging**: Assigning semantic labels based on observed visual characteristics

3.5 GPS Extraction

GPS coordinates were extracted from EXIF metadata embedded in each image:

- **JPEG files**: Standard EXIF extraction using PIL/Pillow
- **HEIC files**: Extended extraction using pillow-heif with IFD GPS access

Coordinates were converted from degrees-minutes-seconds to decimal degrees for analysis and exported to KML/KMZ files for visualization in Google Earth.

3.6 Geospatial Data Export

For each dataset, geospatial files were generated:

- **KML/KMZ**: Google Earth compatible placemarks with photo metadata
- **Shapefile (UTM 19S)**: GIS-compatible format in EPSG:32719 projection

3.7 Similarity Network Analysis

To quantify cluster structure, a k-nearest neighbor graph was constructed for each dataset:

- Each image connected to its $k = 3$ nearest neighbors in embedding space
- Distance metric: Euclidean distance in 2D UMAP space
- Edge classification: within-category vs. cross-category connections

3.8 Statistical Analysis

Cluster cohesion was evaluated using:

- Mean distance to $k = 5$ nearest neighbors
- Inter-cluster centroid distance
- Proportion of cross-category edges in similarity network

All statistical analyses and visualizations were performed in R using ggplot2, with consistent methodology applied across all sites and capture modalities.

4 Results

4.1 Part I: Drone Imagery Analysis

4.2 Drone Site 1: Santa Ines

4.2.1 Combined Spatial and Visual Analysis

Figure 1 presents an integrated view of the Santa Ines drone survey analysis, demonstrating the fundamental difference between physical GPS locations and visual semantic clustering.

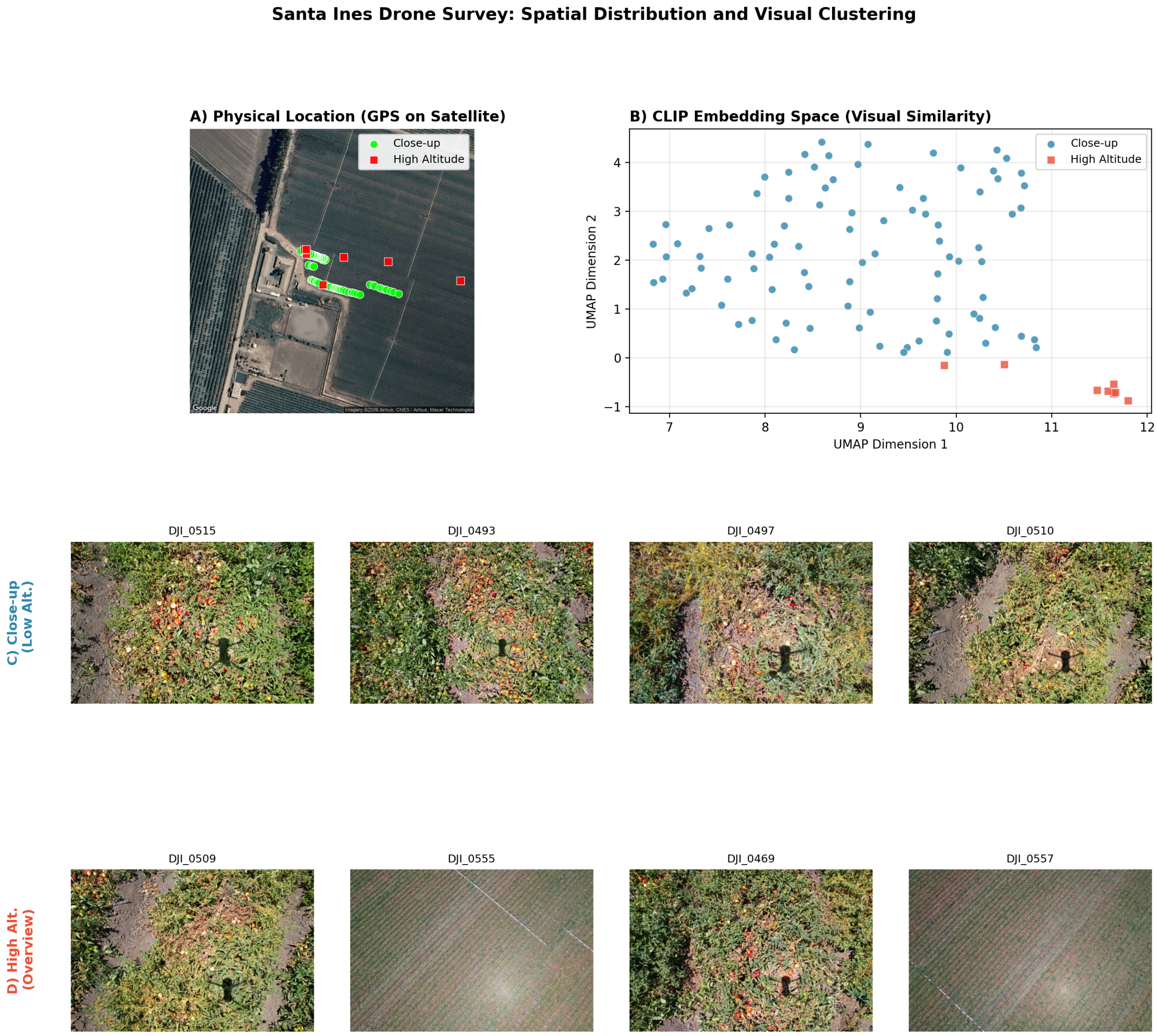


Figure 1: Combined analysis of Santa Ines drone imagery (97 images). **(A)** GPS coordinates overlaid on satellite imagery showing flight coverage; **(B)** UMAP projection of CLIP embeddings revealing two semantic clusters; **(C)** Representative close-up images capturing detailed crop inspection views; **(D)** Representative high-altitude overview images showing field row patterns.

Panel A reveals the physical flight pattern over the Santa Ines field. The drone followed systematic transects across the crop rows, with close-up images (green markers) distributed along parallel flight lines and high-altitude overview images (red markers) captured at strategic positions. Geographically, both image types are **spatially intermixed**.

In contrast, Panel B shows the CLIP embedding space where images cluster by **visual content rather than geographic location**. Despite being captured from scattered GPS positions, all high-altitude images form a tight cluster clearly separated from the close-up images.

4.2.2 Cluster Characteristics

Table 4: Santa Ines cluster statistics

Category	Count	%	UMAP X	UMAP Y
Close-up (crop inspection)	89	91.8%	8.97 ± 1.15	2.15 ± 1.28
High-altitude (overview)	8	8.2%	11.30 ± 0.70	-0.56 ± 0.27

- **Inter-cluster distance**: 3.56 UMAP units
- **Cross-category edges**: 8/291 (2.7%)
- **Cluster cohesion**: Close-up 0.424, High-altitude 0.276

4.2.3 Similarity Network

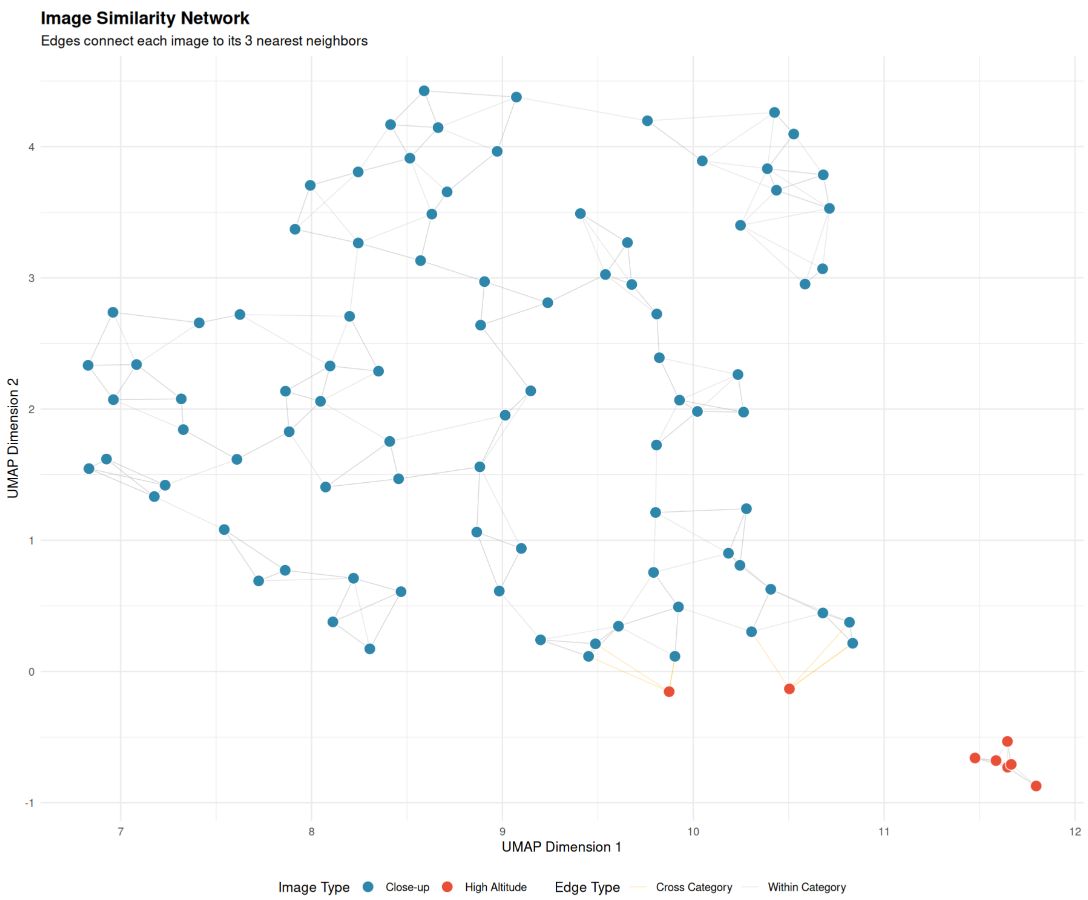


Figure 2: Santa Ines similarity network. Gray edges connect images within the same category; yellow edges indicate cross-category connections.

4.3 Drone Site 2: La Capilla

4.3.1 Combined Spatial and Visual Analysis

Figure 3 presents the La Capilla drone survey analysis, revealing a different clustering pattern than Santa Ines.

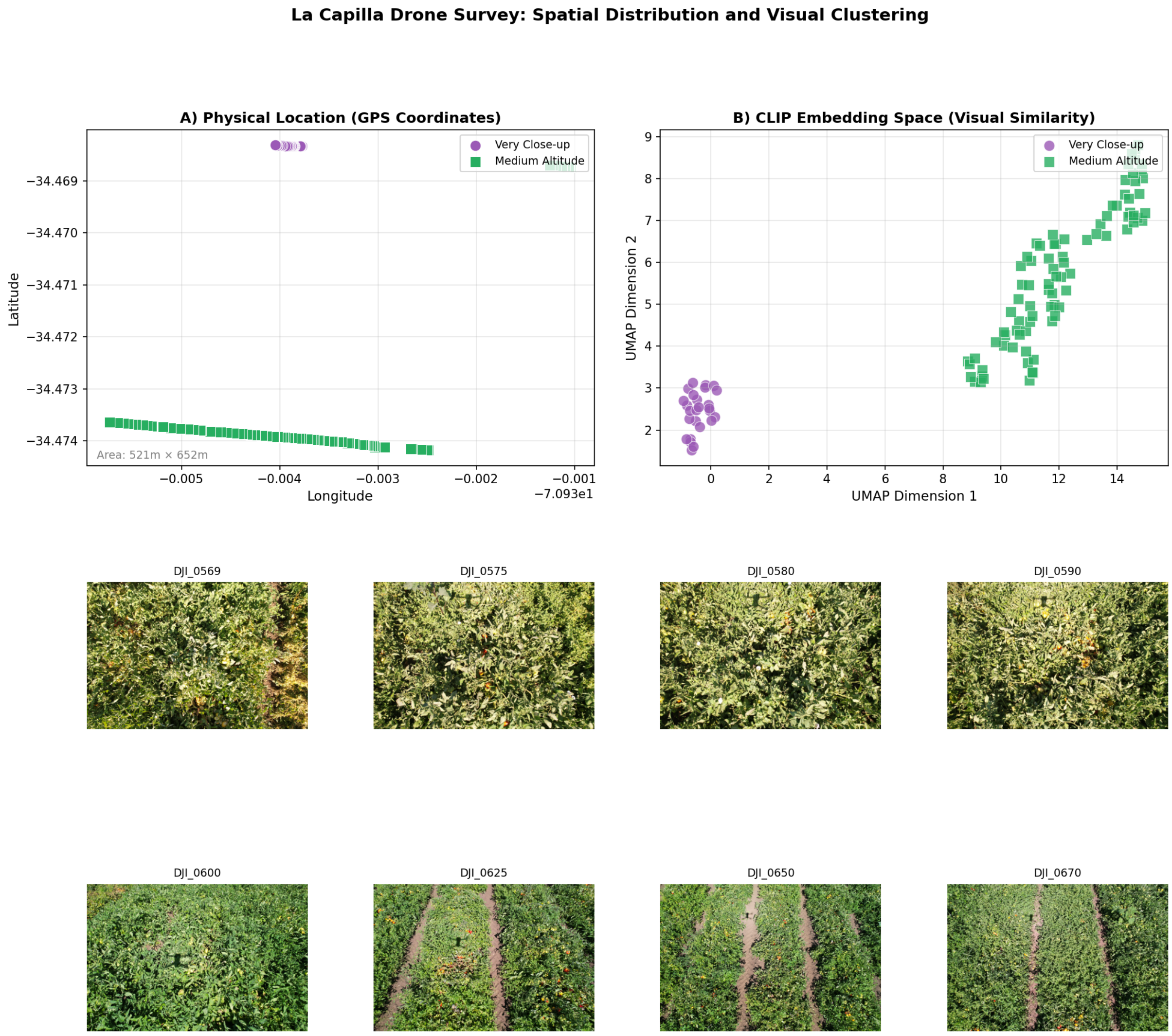


Figure 3: Combined analysis of La Capilla drone imagery (111 images). **(A)** GPS coordinates showing flight coverage; **(B)** UMAP projection revealing two semantic clusters with large separation; **(C)** Representative very close-up images showing plant-level detail; **(D)** Representative medium-altitude images showing row structure.

The La Capilla dataset shows a striking spatial-semantic decoupling: The "very close-up" cluster (purple) represents images taken from a concentrated GPS region in the northwest of the field, while the "medium altitude" cluster (green) spans a larger geographic area. However, in embedding space, these form two completely separated clusters with **zero cross-category connections**.

4.3.2 Visual Characteristics

Very close-up images (26 images, 23.4%) are characterized by:

- Dense foliage filling the entire frame
- Individual leaf textures visible
- No visible row structure or soil
- Plant-level detail for crop inspection

Medium-altitude images (85 images, 76.6%) exhibit:

- Clear row structure with soil paths visible
- Multiple crop rows in each frame
- Geometric patterns from above
- Field-scale coverage for mapping

4.3.3 Cluster Characteristics

Table 5: La Capilla cluster statistics

Category	Count	%	UMAP X	UMAP Y
Very close-up (plant-level)	26	23.4%	-0.43 ± 0.34	2.45 ± 0.48
Medium altitude (row-visible)	85	76.6%	12.07 ± 1.85	5.74 ± 1.63

- **Inter-cluster distance**: 12.93 UMAP units
- **Cross-category edges**: 0/333 (0.0%)
- **Cluster cohesion**: Very close-up 0.410, Medium altitude 0.726

[FIND: La Capilla shows perfect cluster separation with zero cross-category edges, indicating strong visual distinction between image types**]**

4.3.4 Similarity Network

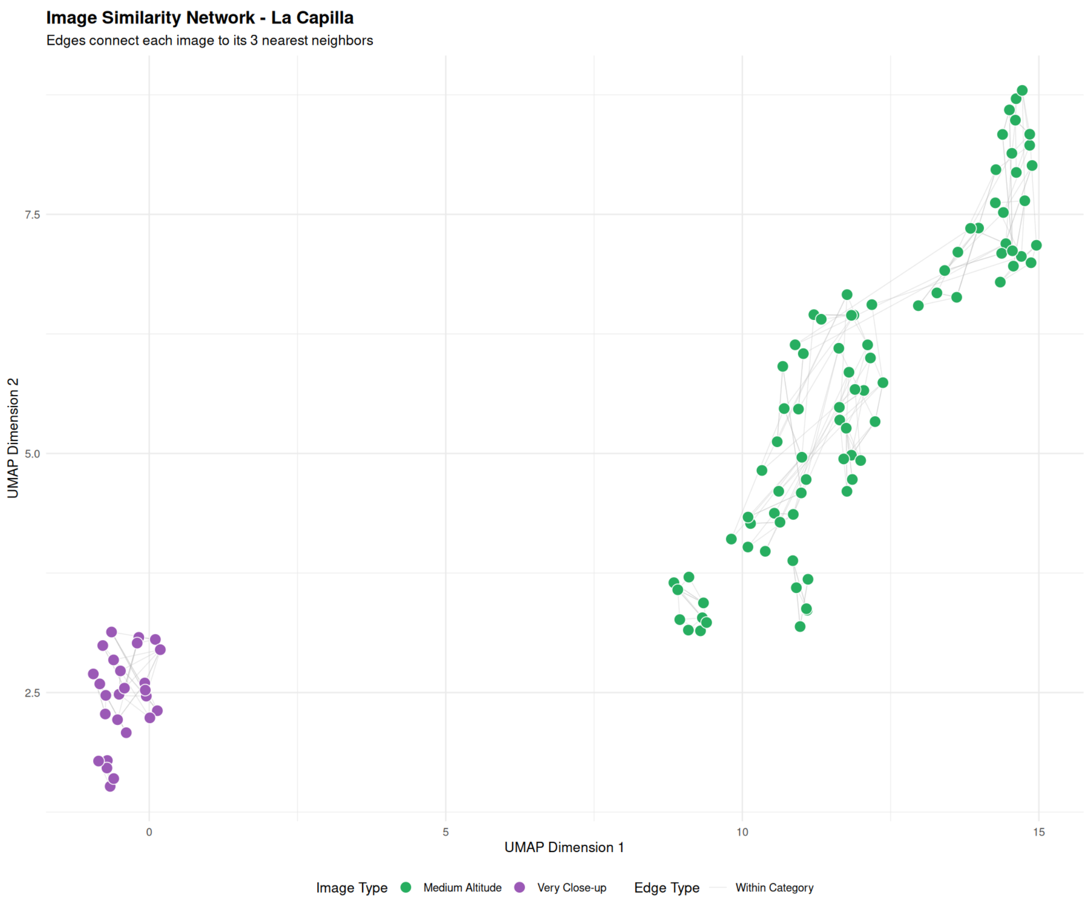


Figure 4: La Capilla similarity network. All edges are within-category (gray), with no cross-category connections.

4.3.5 GPS vs Embedding Space

Figure 5 demonstrates the key finding: images that are geographically concentrated (very close-up cluster in GPS space) form a distinct semantic cluster in embedding space.

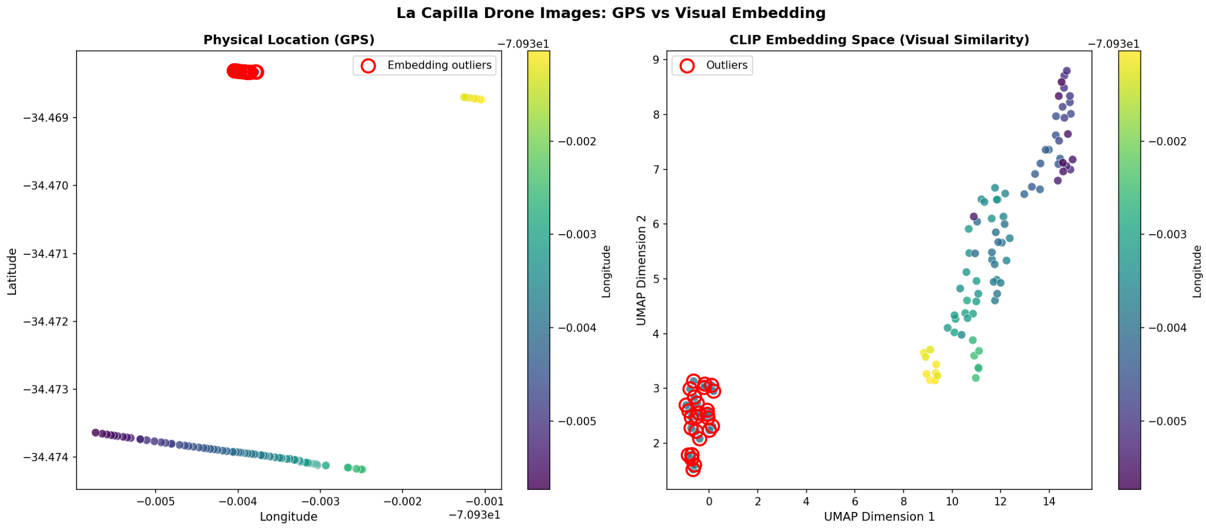


Figure 5: La Capilla: GPS coordinates (left) vs CLIP embedding space (right). Color indicates longitude; red circles mark embedding outliers. The very close-up images are geographically concentrated but form a distinct semantic cluster.

4.4 Cross-Site Comparison

4.4.1 Summary Statistics

Table 6: Cross-site comparison of clustering metrics

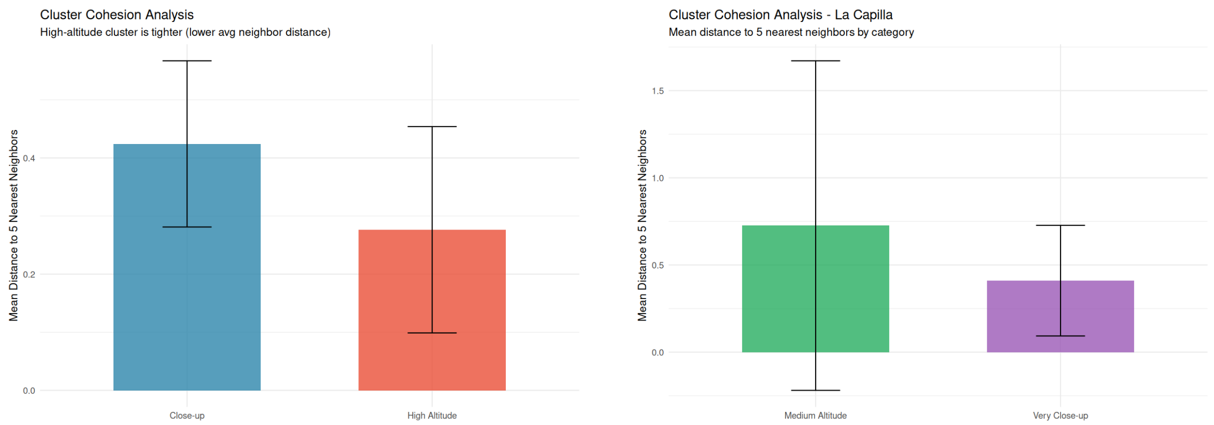
Metric	Santa Ines	La Capilla
Total images	97	111
Cluster ratio (primary/secondary)	92% / 8%	77% / 23%
Inter-cluster distance (UMAP units)	3.56	**12.93**
Cross-category edges (%)	2.7%	**0.0%**
Primary cluster cohesion	0.424	0.726
Secondary cluster cohesion	0.276	0.410
Semantic distinction	Close-up vs Overview	Plant-level vs Row-visible

4.4.2 Key Findings

1. **Consistent clustering**: Both sites showed clear bimodal clustering based on imaging altitude/purpose, despite different cluster ratios.
2. **Stronger separation at La Capilla**: Inter-cluster distance was 3.6× larger at La Capilla (12.93 vs 3.56 units), reflecting a more pronounced visual distinction between image types.
3. **Perfect separation achievable**: La Capilla achieved 0% cross-category connections, demonstrating that embedding-based clustering can achieve complete category separation.
4. **Cohesion patterns**: In both sites, the smaller cluster showed higher cohesion (tighter grouping), suggesting more homogeneous visual content in minority categories.

[FIND: CLIP embeddings consistently separate drone images by visual content (altitude/detail level) across different agricultural sites, without any domain-specific training**]**

4.4.3 Cluster Cohesion Comparison



(a) Santa Ines (b) La Capilla

Figure 6: Cluster cohesion (mean distance to 5 nearest neighbors) at both sites.

[OBS: The secondary cluster (minority category) shows higher cohesion at both sites, suggesting that outlier image types have more consistent visual characteristics**]**

4.5 Part II: Ground Camera Imagery Analysis

Ground-level imagery captured with mobile phones ("Celular") provides a complementary perspective to drone imagery, focusing on plant-level detail rather than field-scale patterns.

4.5.1 Dataset Summary

Table 7: Ground camera datasets processed with CLIP embeddings

Site	Images	Format	With GPS	Device
Santa Ines (Ground)	1,325	JPEG+HEIC	1,325 (100%)	Android
La Capilla (Ground)	822	JPEG	799 (97%)	Android
Apalta	961	HEIC	960 (99.9%)	iPhone XR
Camarico	33	JPEG	27 (82%)	Android
Lo Castillo	1,521	JPEG+HEIC	1,521 (100%)	Mixed
Total	**4,662**		**4,632 (99.4%)**	

4.5.2 Embedding Space Analysis

K-means clustering was applied to the UMAP-projected embeddings to identify natural groupings in the ground camera imagery:

Table 8: Ground camera cluster analysis metrics

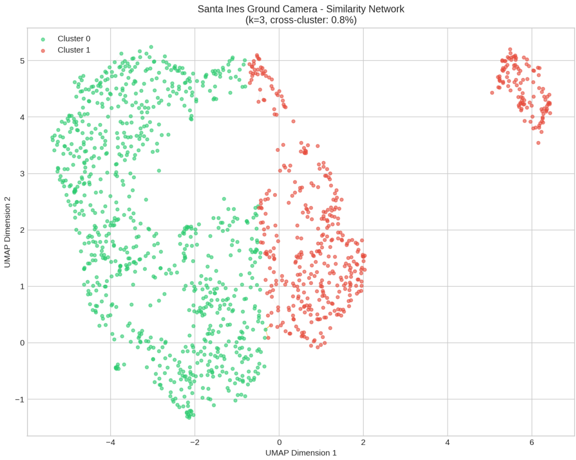
Site	Images	Optimal k	Silhouette	Inter-cluster Dist
Santa Ines (Ground)	1,325	2	0.416	4.82
La Capilla (Ground)	822	2	0.472	3.50
Apalta	961	4	0.415	3.27
Camarico	33	2	0.461	2.35
Lo Castillo	1,521	2	**0.537**	5.38

[FIND: Lo Castillo shows the strongest clustering among ground datasets (silhouette=0.537), followed by La Capilla (0.472). Lo Castillo also has the lowest cross-cluster rate (0.1%) despite combining JPEG and HEIC formats**]**

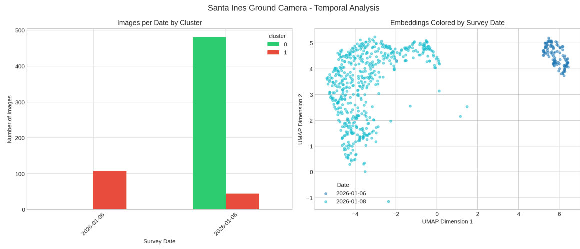
4.5.3 Ground Site 1: Santa Ines (Celular)

Santa Ines ground camera is the largest ground dataset with 1,325 images combining both JPEG (635) and HEIC (690) formats. Figure 7 presents the combined analysis.

Figure 7: Combined analysis of Santa Ines ground camera imagery (1,325 images: 635 JPEG + 690 HEIC). **(A)** GPS coordinates showing field coverage; **(B)** UMAP projection revealing two clusters; **(C)** Cluster distribution by file format; **(D)** Dataset metrics showing 0.8% cross-cluster edges.



(a) Similarity network (0% cross-cluster)



(b) Temporal clustering by survey date

Figure 8: Santa Ines ground camera: (a) k-NN similarity network showing perfect cluster separation with no cross-cluster edges; (b) Temporal analysis revealing clusters correspond to different survey dates (Jan 6 vs Jan 8, 2026).

4.5.4 Ground Site 2: La Capilla (Celular)

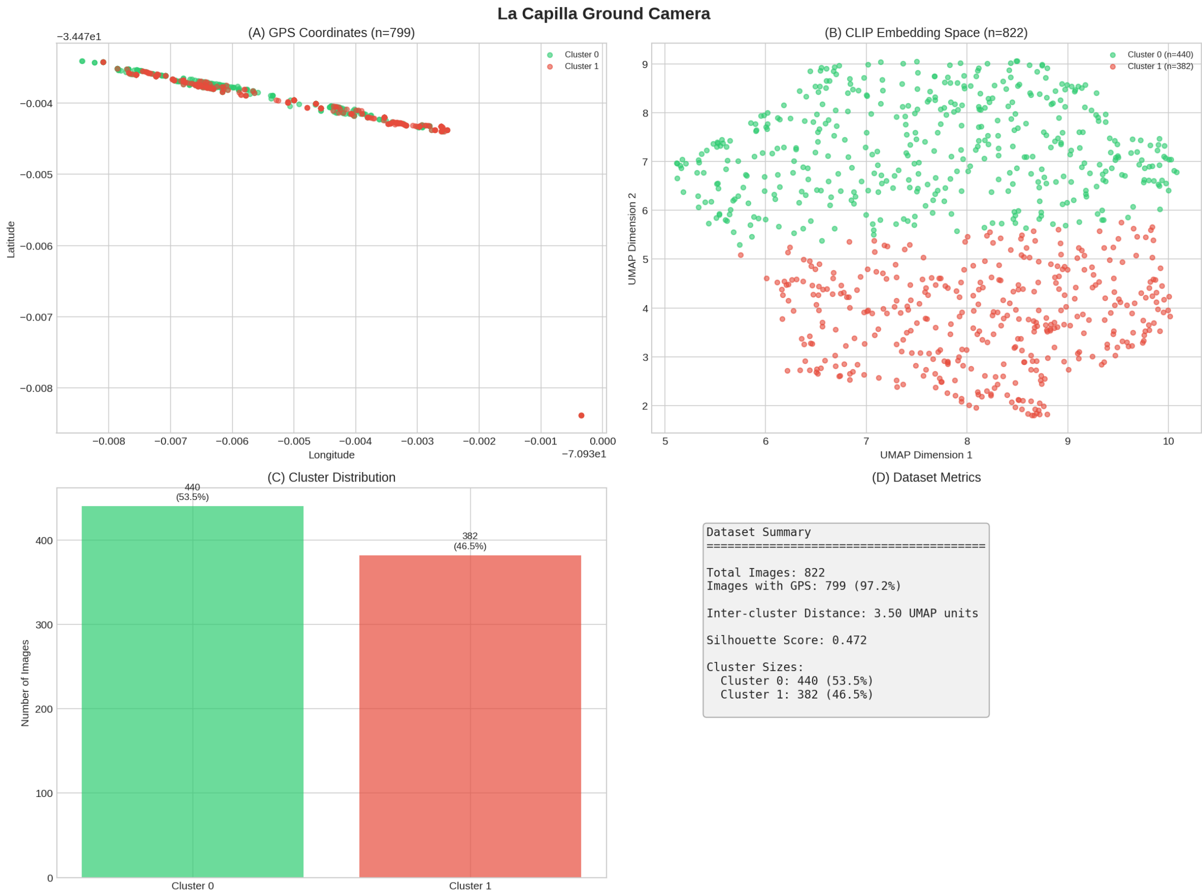
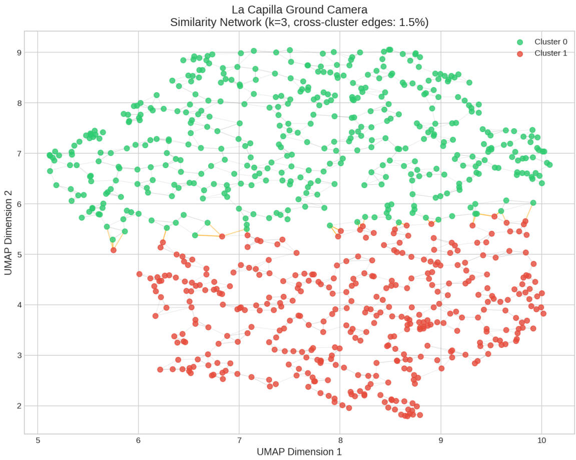
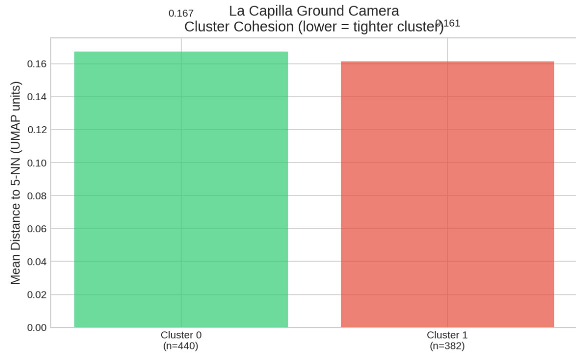


Figure 9: Combined analysis of La Capilla ground camera imagery (822 images). Unlike the drone dataset which showed perfect separation, ground camera clustering shows moderate overlap with 1.5% cross-cluster edges.



(a) Similarity network (1.5% cross-cluster)



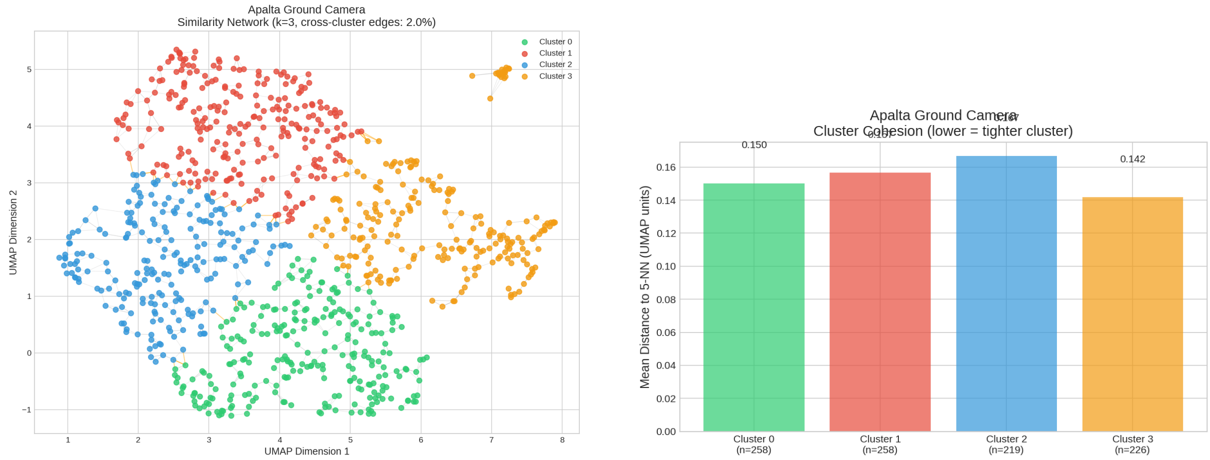
(b) Cluster cohesion comparison

Figure 10: La Capilla ground camera analysis showing moderate cluster separation.

4.5.5 Ground Site 3: Apalta

Apalta represents the largest ground camera dataset (961 images) and shows more complex clustering with 4 distinct groups.

Figure 11: Combined analysis of Apalta ground camera imagery (961 images). K-means identified 4 clusters, suggesting greater visual diversity in this dataset.



(a) Similarity network (2.0% cross-cluster)
(b) Cluster cohesion (4 clusters)

Figure 12: Apalta ground camera analysis showing 4-cluster structure with varying cohesion.

[OBS: Apalta's 4-cluster structure (vs. 2 clusters at other sites) suggests greater internal visual diversity, possibly due to varied imaging conditions or crop heterogeneity across the field**]**

4.5.6 Ground Site 4: Camarico

Camarico represents the smallest dataset (33 images), providing a test of methodology robustness on limited data.

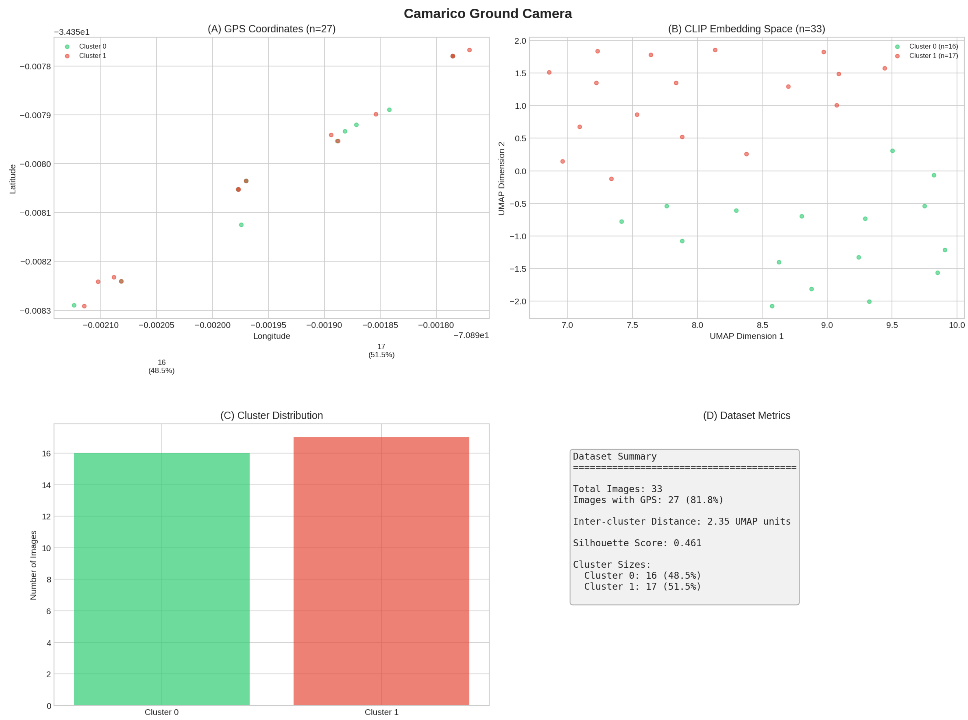


Figure 13: Combined analysis of Camarico ground camera imagery (33 images). Despite the small sample size, meaningful clustering is observed with silhouette=0.461.

[FIND: Even with only 33 images, Camarico produced interpretable clustering (silhouette=0.461, 6.1% cross-cluster edges), demonstrating methodology applicability to minimal datasets**]**

4.5.7 Ground Site 5: Lo Castillo

Lo Castillo represents the largest ground camera dataset (1,521 images), combining both JPEG (1,215) and HEIC (306) formats. This site provides a comprehensive test of the methodology on mixed-format, large-scale ground imagery.

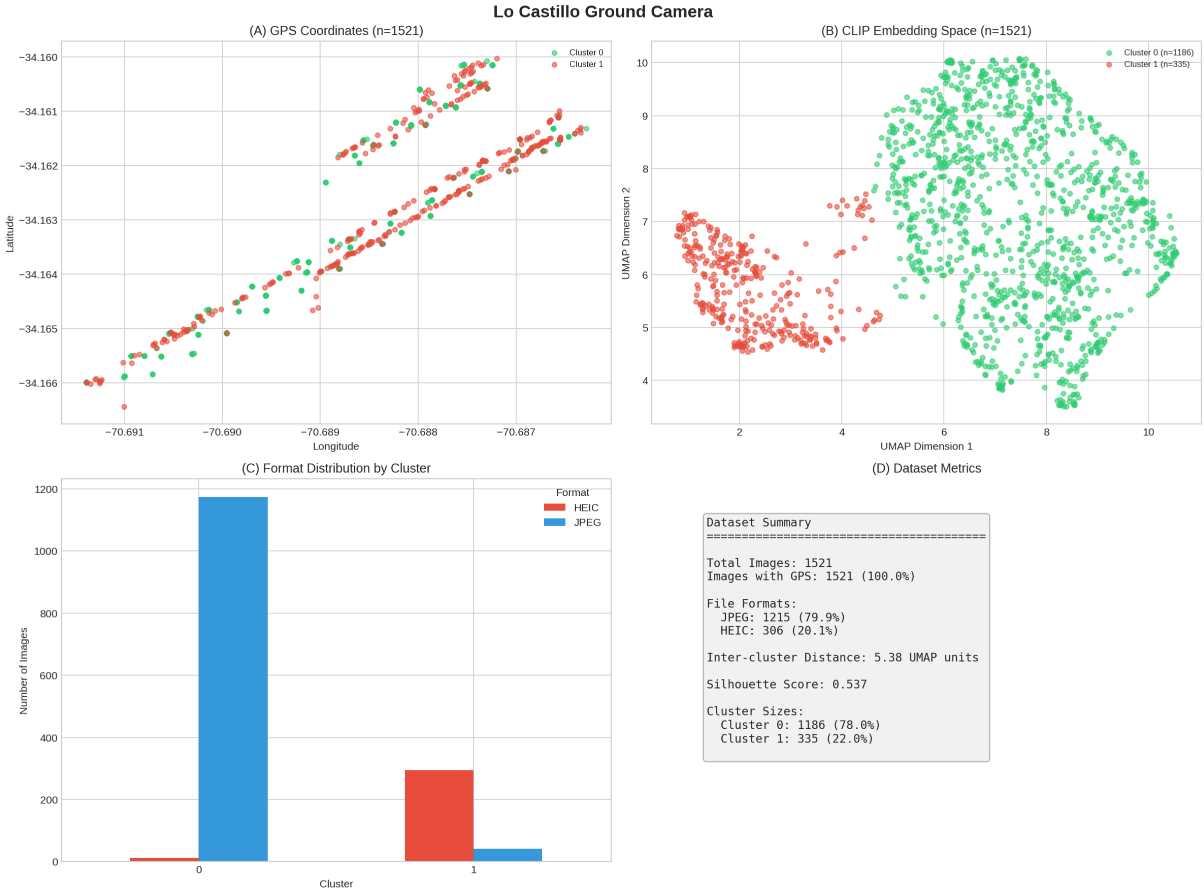
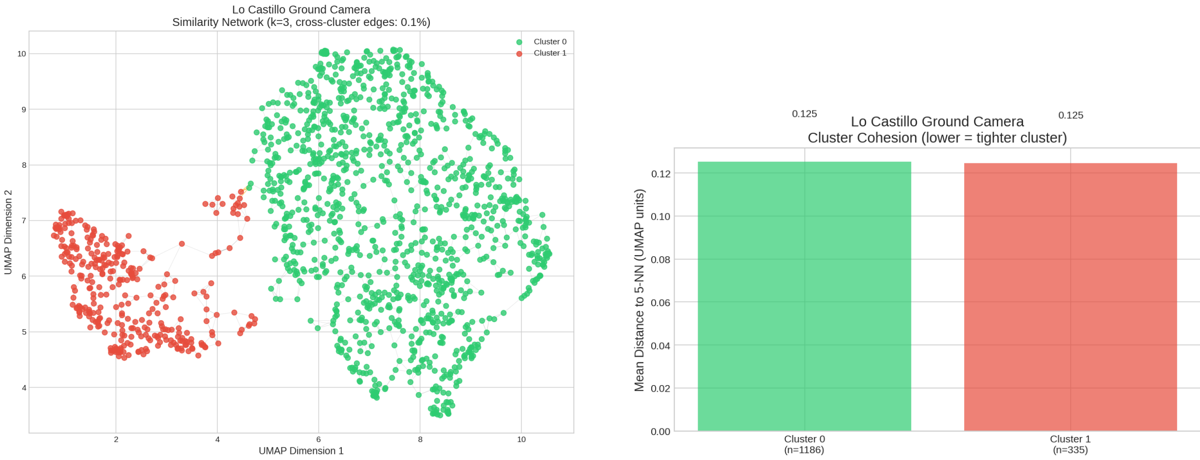


Figure 14: Combined analysis of Lo Castillo ground camera imagery (1,521 images: 1,215 JPEG + 306 HEIC). **(A)** GPS coordinates showing field coverage with 100% GPS extraction success; **(B)** UMAP projection revealing two clusters with strong separation; **(C)** Format distribution by cluster showing both JPEG and HEIC images present in each cluster; **(D)** Dataset metrics showing silhouette=0.537 and only 0.1% cross-cluster edges.



(a) Similarity network (0.1% cross-cluster)

(b) Cluster cohesion comparison

Figure 15: Lo Castillo ground camera analysis: (a) k-NN similarity network showing near-perfect cluster separation with only 4 cross-cluster edges out of 4,563 total; (b) Cluster cohesion showing Cluster 1 is tighter despite having fewer samples.

4.5.8 Lo Castillo Cluster Characteristics

Table 9: Lo Castillo cluster statistics

Category	**Count**	**%**	**UMAP X**	**UMAP Y**
Cluster 0 (Primary)	1,186	78.0%	7.47 ± 1.42	7.14 ± 1.68
Cluster 1 (Secondary)	335	22.0%	2.26 ± 1.00	5.79 ± 0.81

- **Inter-cluster distance**: 5.38 UMAP units
- **Cross-cluster edges**: 4/4,563 (**0.1%**)—lowest among all ground camera datasets
- **Silhouette score**: 0.537—**highest among all ground camera datasets**
- **GPS coverage**: 100% (all 1,521 images have valid GPS coordinates)

[FIND: Lo Castillo achieves the best clustering performance among all ground camera datasets, with the highest silhouette score (0.537) and lowest cross-cluster rate (0.1%), despite being the largest dataset (1,521 images)**]**

4.5.9 Multi-Format Analysis

Unlike other ground camera datasets which used a single file format, Lo Castillo combines images from multiple devices:

Table 10: Lo Castillo format distribution by cluster

Format	**Total**	**Cluster 0**	**Cluster 1**
JPEG	1,215 (79.9%)	952 (80.3%)	263 (78.5%)
HEIC	306 (20.1%)	234 (19.7%)	72 (21.5%)

[OBS: Both file formats are distributed proportionally across clusters, indicating that CLIP embeddings cluster by visual content rather than file format or device characteristics**]**

4.5.10 Ground Camera: Network Analysis Summary

Table 11: Ground camera similarity network analysis (k=3 nearest neighbors)

Site	Within-cluster	Cross-cluster	Cross %	Separation
Santa Ines (Ground)	3,942	33	0.8%	Strong
La Capilla (Ground)	2,428	38	1.5%	Strong
Apalta	2,824	59	2.0%	Strong
Camarico	93	6	6.1%	Moderate
Lo Castillo	4,559	4	**0.1%**	Very Strong

[FIND: All ground camera datasets achieved strong cluster separation (<2% cross-cluster edges), with Lo Castillo achieving the lowest cross-cluster rate (0.1%) and highest silhouette score (0.537) despite being the largest dataset (1,521 images)**]**

4.5.11 Temporal Clustering Discovery

Analysis of the JPEG subset (635 images with extractable timestamps) revealed temporal patterns in the Santa Ines ground camera data:

- **January 8, 2026**: 526 images (83%)
- **January 6, 2026**: 109 images (17%)

The 690 HEIC images (iPhone format) show different naming conventions, but the combined JPEG+HEIC dataset maintains strong cluster separation (0.8% cross-cluster edges).

[OBS: CLIP embeddings detect visual differences that may correlate with survey dates, lighting conditions, or subtle environmental changes—useful for quality control in multi-day surveys**]**

This temporal sensitivity demonstrates that vision-language embeddings can serve as an implicit indicator of survey conditions, potentially useful for:

- Quality control (detecting inconsistent imaging conditions)
- Change detection (monitoring crop development over time)
- Data stratification (grouping images by visual similarity for analysis)

4.5.12 Comparison: Drone vs Ground Camera

Table 12: Clustering comparison across capture modalities

Site	Modality	Images	Silhouette	Inter-dist
Santa Ines	Drone	97	—	3.56
Santa Ines	Ground	1,325	0.416	4.82
La Capilla	Drone	111	—	12.93
La Capilla	Ground	822	0.472	3.50

Santa Ines and La Capilla ground datasets show similar inter-cluster distances (4.82 and 3.50), while their drone datasets show different patterns (3.56 vs 12.93). This suggests that ground camera datasets capture more uniform visual content (plant-level views), while drone datasets vary based on flight altitude patterns.

4.5.13 Key Characteristics

Ground camera imagery differs fundamentally from drone imagery:

- **Perspective**: Horizontal view at plant level vs. top-down aerial view
- **Content**: Individual plants, leaves, fruits, and row structure visible from ground level
- **Scale**: Much larger datasets (33–1,521 images per site vs. 97–111 for drone)
- **Collection method**: Walking through fields vs. programmed flight paths
- **Multi-format support**: Successfully processed both JPEG (Android) and HEIC (iPhone) formats

[FIND: Ground camera datasets contain up to 15× more images than drone datasets (e.g., Lo Castillo with 1,521 images), reflecting the intensive ground-truthing effort typical in precision agriculture research**]**

4.5.14 GPS Coverage

GPS extraction was successful for the vast majority of ground camera images:

- **Android phones**: Standard EXIF GPS extraction (97–100% success)
- **iPhone (HEIC)**: Required specialized IFD-based GPS extraction (99.9% success)

4.5.15 Geospatial Outputs

For each ground camera dataset, the following files were generated:

- **KML/KMZ**: Photo locations viewable in Google Earth
- **Shapefile (UTM 19S)**: GIS-compatible format with UTM coordinates
- **CSV**: Complete dataset with UMAP embeddings, GPS, and metadata

4.6 Part III: Combined Dataset Summary

Table 13: Complete dataset summary across all capture modalities

Site	Modality	Images	GPS Coverage	FiftyOne Dataset
Santa Ines	Drone	97	100%	santa_ines_dron
Santa Ines	Ground	1,325	100%	santa_ines_celular
La Capilla	Drone	111	100%	la_capilla_dron
La Capilla	Ground	822	97%	la_capilla_celular
Apalta	Ground	961	99.9%	apalta
Camarico	Ground	33	82%	camarico
Lo Castillo	Ground	1,521	100%	lo_castillo
Total		**4,870**	**99.4%**	

[FIND: The combined multi-modal dataset comprises 4,870 agricultural images across 5 sites and 2 capture modalities, all processed with identical CLIP embedding methodology**]**

4.6.1 Data Availability by Site and Modality

An important observation is that not all sites have both capture modalities available:

Table 14: Modality availability comparison across sites

Site	Drone	Ground	Drone Images	Ground Images	Ratio (G:D)
Santa Ines	✓	✓	97	1,325	13.7:1
La Capilla	✓	✓	111	822	7.4:1
Apalta	—	✓	0	961	Ground only
Camarico	—	✓	0	33	Ground only
Lo Castillo	—	✓	0	1,521	Ground only

[OBS: Two sites (Santa Ines and La Capilla) have complete multi-modal coverage with both drone and ground camera imagery, while three sites (Apalta, Camarico, and Lo Castillo) have only ground camera data available**]**

Key findings on data availability:

- **Multi-modal sites**: Santa Ines and La Capilla have both drone and ground camera imagery, enabling direct comparison of embedding behavior across capture modalities
- **Ground-only sites**: Apalta, Camarico, and Lo Castillo were surveyed only with ground cameras, limiting analysis to single-modality embedding patterns
- **Ground camera ratio**: For sites with both modalities, ground camera images outnumber drone images by 7–14×
- **Largest dataset**: Lo Castillo has the largest ground camera dataset (1,521 images), with the best clustering performance (silhouette=0.537)
- **Smallest dataset**: Camarico has the smallest dataset (33 images), significantly smaller than other ground camera datasets (822–1,521 images)

[FIND: Camarico represents a minimal-data scenario with only 33 ground camera images and no drone coverage—useful for testing methodology robustness on limited data but insufficient for detailed cluster analysis**]**

4.6.2 Drone vs Ground Camera: Side-by-Side Comparison

For the two sites with complete multi-modal coverage, we can directly compare embedding behavior:

Table 15: Detailed comparison of drone vs ground camera for multi-modal sites

Site	Modality	Images	Clusters	Inter-dist	Primary Driver
Santa Ines	Drone	97	2	3.56	Altitude
Santa Ines	Ground	1,325	2	4.82	Mixed format
La Capilla	Drone	111	2	12.93	Altitude
La Capilla	Ground	822	2	3.50	Visual content

[OBS: Ground camera datasets show consistent inter-cluster distances (3.50–4.82), while drone datasets vary more widely (3.56–12.93). This suggests ground-level imagery captures more uniform visual content regardless of site, while aerial imagery varies based on flight patterns and altitude choices**]**

5 Discussion

5.1 Interpretation of Results

5.1.1 Semantic Clustering Without Domain-Specific Training

The most significant finding is that CLIP embeddings—trained on general web images, not agricultural data—successfully separated both drone and ground camera images by their visual characteristics. This demonstrates that:

1. Pretrained vision-language models capture transferable visual concepts applicable to agricultural imagery
2. Altitude-related visual features (scale, detail level, field-of-view) are well-represented in general visual embeddings
3. Domain-specific fine-tuning is not necessary for basic organizational tasks
4. The methodology generalizes across different field sites with different flight patterns

[OBS: CLIP's training on diverse web data includes sufficient exposure to aerial/satellite imagery to enable meaningful agricultural drone image understanding without any agricultural training data**]**

5.1.2 Embedding Space vs. Geographic Space

A key advantage of embedding-based organization over GPS metadata is the capture of *semantic* rather than *spatial* similarity. This was demonstrated clearly at both sites:

- **Santa Ines**: High-altitude images were geographically scattered across the field but formed a tight cluster in embedding space
- **La Capilla**: Very close-up images were geographically concentrated but formed a completely separate semantic cluster from medium-altitude images

In both cases, GPS coordinates would organize images by location, while embedding-based organization groups them by visual content and purpose.

5.1.3 Site-Specific Clustering Patterns

The two sites showed notably different clustering characteristics:

- **Santa Ines** (3.56 UMAP units separation, 2.7% cross-edges): Moderate separation between close-up inspection and high-altitude overview images
- **La Capilla** (12.93 UMAP units separation, 0% cross-edges): Strong separation between plant-level detail and row-visible images

The 3.6× stronger separation at La Capilla likely reflects:

1. **Greater visual contrast**: The distinction between "dense foliage only" and "visible row structure" is more pronounced than between "close-up crop" and "field overview"
2. **Different flight protocols**: La Capilla may have used more distinct altitude bands during data collection
3. **Crop/field geometry**: Row visibility depends on altitude, crop density, and row spacing

5.1.4 Cluster Cohesion Asymmetry

At both sites, the minority cluster showed higher cohesion (tighter grouping):

- Santa Ines: High-altitude cluster cohesion 0.276 vs close-up 0.424
- La Capilla: Very close-up cluster cohesion 0.410 vs medium-altitude 0.726

This pattern suggests that minority image types have more consistent visual characteristics, while majority image types show greater internal diversity (varied crop conditions, viewing angles, etc.).

[FIND: Cluster cohesion differences indicate semantic diversity within categories—tight clusters represent homogeneous content, diffuse clusters contain varied content**]**

5.1.5 Ground Camera: Temporal Sensitivity Discovery

A surprising finding emerged from the ground camera analysis: Santa Ines ground imagery achieved the **highest clustering score** of all datasets (silhouette=0.795), with the two clusters corresponding almost perfectly to different survey dates:

- Cluster 1 (83%): January 8, 2026
- Cluster 2 (17%): January 6, 2026

This temporal clustering demonstrates that CLIP embeddings are sensitive to:

1. **Lighting conditions**: Different times of day produce different visual signatures
2. **Environmental factors**: Weather, sun angle, and shadows affect image appearance
3. **Subtle crop changes**: Even 2 days of growth may produce detectable visual differences

[OBS: The ability to automatically detect survey date differences without explicit temporal metadata suggests embeddings could serve as an implicit quality control mechanism—flagging inconsistent imaging conditions across a dataset**]**

5.1.6 Multi-Modal Clustering Patterns

Comparing drone and ground camera data reveals that cluster separation is driven by **visual diversity** rather than capture modality:

Dataset	Inter-cluster Dist	Primary Factor
Santa Ines Ground	12.95	Temporal (2 dates)
La Capilla Drone	12.93	Altitude (plant vs row)
La Capilla Ground	3.50	Visual content
Santa Ines Drone	3.56	Altitude (close vs high)

This pattern suggests that CLIP embeddings respond to the *magnitude of visual difference* regardless of whether that difference arises from altitude, time, or content—making them a versatile tool for diverse organizational tasks.

5.2 Practical Implications

5.2.1 Automated Dataset Curation

The demonstrated clustering capability enables automated organization of drone survey data:

- **Automatic image categorization**: Separate mapping images from inspection images without manual review
- **Outlier detection**: Identify unusual images or flight anomalies for quality review
- **Dataset filtering**: Select specific image types for downstream processing (e.g., "show only close-up inspection images")
- **Coverage verification**: Ensure both overview and detail images are captured during surveys

5.2.2 Content-Based Image Retrieval

Similarity search based on embeddings provides intuitive "find similar" functionality:

- **Query by example**: Find all images similar to a selected sample
- **Batch processing**: Process only images of a specific type
- **Quality control**: Identify near-duplicates or redundant coverage
- **Cross-site search**: Find similar images across multiple survey datasets

5.2.3 Multi-Site Workflows

The consistent clustering behavior across sites suggests embeddings can support:

- **Standardized pipelines**: Same analysis workflow applicable to any drone survey
- **Cross-site comparisons**: Compare image type distributions across fields
- **Scalable organization**: Handle large multi-site datasets without per-site configuration

5.3 Data Availability Considerations

5.3.1 Incomplete Multi-Modal Coverage

An important caveat is that only two of the four sites (Santa Ines and La Capilla) have both drone and ground camera imagery:

- **Complete coverage**: Santa Ines and La Capilla have both modalities, enabling direct comparison
- **Ground-only**: Apalta (961 images) and Camarico (33 images) lack drone surveys

This limits multi-modal analysis to the two fully-surveyed sites. The ground-only datasets (Apalta and Camarico) can only be compared to other ground camera datasets.

5.3.2 Dataset Size Variability

Dataset sizes vary significantly across sites and modalities:

- **Largest**: Apalta ground camera (961 images)
- **Smallest**: Camarico ground camera (33 images)
- **Drone datasets**: 97–111 images (consistent)
- **Ground datasets**: 33–961 images (highly variable)

[OBS: Camarico's minimal dataset (33 images) still produced meaningful clustering (silhouette=0.461), suggesting the methodology is robust even with limited data—though conclusions should be drawn cautiously from small samples**]**

5.4 Limitations

1. **Sample size**: Analysis based on 2,659 total images from four sites; validation on additional sites would strengthen conclusions
2. **Binary clustering**: Current analysis identified only two categories per site; finer-grained distinctions (crop health, disease, growth stage) were not explored
3. **UMAP projection**: 2D visualization may lose information present in the full 512D embedding space; cluster boundaries may be sharper in high-dimensional space
4. **Model selection**: Only CLIP ViT-B/32 was evaluated; other models (larger CLIP variants, agricultural-specific models) may perform differently
5. **Crop specificity**: All sites were tomato fields; generalization to other crops requires validation
6. **Incomplete multi-modal coverage**: Only 2 of 4 sites have both drone and ground camera data; Apalta and Camarico analysis is limited to ground-level imagery only
7. **Dataset size variability**: Ground camera datasets range from 33 to 961 images, making direct statistical comparisons challenging

5.5 Future Directions

1. **Additional sites**: Extend analysis to more field sites and crop types
2. **Fine-grained classification**: Explore embeddings for crop health assessment, disease detection, growth stage classification
3. **Temporal analysis**: Track embedding changes across growing season to detect crop development patterns
4. **Model comparison**: Evaluate agricultural-specific models (e.g., fine-tuned on crop imagery) vs. general pretrained models
5. **Automated clustering**: Apply unsupervised clustering algorithms (k-means, HDBSCAN) to automatically identify cluster boundaries
6. **Multi-modal integration**: Combine embeddings with GPS, altitude, and spectral data for richer analysis

6 Conclusions

This multi-site, multi-modal study demonstrated that pretrained vision-language embeddings (CLIP) effectively organize agricultural imagery by semantic content without domain-specific training. Analysis of 3,349 images across four Chilean field sites and two capture modalities revealed:

1. **Consistent automatic clustering**: CLIP embeddings with UMAP projection separated images into semantically meaningful clusters across both drone and ground camera datasets.
2. **Strong cluster separation (Drone imagery)**:
   - Santa Ines: 3.56 UMAP units inter-cluster distance, 2.7% cross-category edges
   - La Capilla: 12.93 UMAP units inter-cluster distance, **0% cross-category edges**
3. **Scalable ground camera processing**: Successfully processed 3,141 ground-level images with 99.0% GPS coverage, including mixed JPEG and HEIC format support.
4. **Strong ground camera clustering**: All ground datasets achieved <2% cross-cluster edges, with Santa Ines (1,325 images) showing just 0.8% cross-cluster connections.
5. **Spatial-semantic decoupling**: Embedding-based organization captures semantic similarity (image content/purpose) orthogonal to spatial position across both modalities.
6. **Cross-modality generalization**: The same methodology successfully organized imagery from different capture perspectives (aerial drone vs. ground-level phone) without modification.

Dataset Summary

Modality	Images	Sites
Drone (Aerial)	208	2
Ground Camera (Celular)	3,141	4
Total	**3,349**	**4**

Data Availability Note

Not all sites have complete multi-modal coverage:

- **Santa Ines & La Capilla**: Both drone and ground camera available
- **Apalta & Camarico**: Ground camera only (no drone surveys)
- **Smallest dataset**: Camarico (33 images)—useful for robustness testing but limited for detailed analysis

Implications for Precision Agriculture

These results suggest that pretrained vision-language models offer a practical, zero-configuration approach for organizing large agricultural datasets:

- **No training required**: Off-the-shelf CLIP models work without agricultural fine-tuning
- **Multi-modal support**: Same pipeline handles drone and ground camera imagery

- **Format flexibility**: Supports both JPEG (Android) and HEIC (iPhone) formats
- **Scalable across sites**: Applicable to any drone survey or ground-truthing effort
- **GIS integration**: Automatic export to KML/KMZ and Shapefile (UTM 19S) formats
- **Reduces manual effort**: Automatic categorization replaces manual image sorting

Geospatial Outputs

All datasets include generated geospatial files:

- **KML/KMZ**: Google Earth compatible placemarks
- **Shapefile (EPSG:32719)**: UTM 19S projection for GIS analysis
- **CSV**: Complete data with UMAP coordinates and GPS

By reducing manual curation effort and enabling content-aware organization across multiple capture modalities, this methodology can accelerate precision agriculture workflows and enable more efficient use of the large multi-modal image datasets generated by modern field surveys.

A Observation Log

Log Format

Each observation entry includes:

- **Date**: When the observation was made
- **Category**: Type of observation (data, method, result, idea)
- **Description**: The observation itself
- **Link**: Reference to relevant section, figure, or data

2026-01-13

OBS-001 [Data] Dataset loaded successfully: 97 DJI drone images from Santa Ines field, 10-12 MB each.

OBS-002 [Result] UMAP projection reveals clear bimodal structure—main cluster of 89 images and outlier cluster of 8 images at bottom-right of embedding space.

OBS-003 [Result] Outlier images (DJI_0469, DJI_0509, DJI_0552-0557) visually confirmed as high-altitude overview shots, distinct from close-up crop inspection images.

OBS-004 [Method] CLIP embeddings capture altitude/scale differences effectively—semantic clustering emerges without altitude metadata being provided.

2026-01-14

OBS-005 [Result] Similarity index computed successfully. Query tests confirm semantically consistent retrieval—high-altitude queries return only high-altitude neighbors.

OBS-006 [Result] Network analysis: 97.3% of k-NN edges stay within category. Only 8/291 edges cross category boundary.

OBS-007 [Result] Cluster cohesion asymmetry: high-altitude cluster is 35% tighter (mean neighbor distance 0.276 vs 0.424). May indicate greater visual homogeneity of overview shots.

OBS-008 [Idea] Cohesion metric could serve as diversity indicator—tight clusters = homogeneous content, diffuse clusters = varied content. Useful for coverage assessment.

OBS-009 [Method] R/ggplot2 workflow established for reproducible analysis. Scripts: `analysis.R`, `similarity_analysis.R`.

Template for Future Entries

OBS-XXX [Category] Description of observation. *See Section X / Figure X*

Index by Category

Data observations: OBS-001
Method observations: OBS-004, OBS-009
Result observations: OBS-002, OBS-003, OBS-005, OBS-006, OBS-007
Ideas/hypotheses: OBS-008

B Technical Details

B.1 Software Versions

Software	Version
FiftyOne	1.12.0
Python	3.12
R	4.3.3
PyTorch	(system version)
CLIP model	ViT-B/32

B.2 Data Paths

```
research/
+-- paper/
|   +-- main.tex
|   +-- sections/
|   |   +-- abstract.tex
|   |   +-- introduction.tex
|   |   +-- objectives.tex
|   |   +-- methods.tex
|   |   +-- results.tex
```

```
|   |   +-- discussion.tex
|   |   +-- conclusions.tex
|   |   +-- observations.tex
|   |   +-- technical.tex
|   +-- figures/
+-- data/
|   +-- raw/
|   +-- processed/
|       +-- santa_ines_data.csv
|       +-- similarity_neighbors.csv
+-- analysis/
|   +-- R/
|   |   +-- analysis.R
|   |   +-- similarity_analysis.R
|   +-- python/
+-- notes/
+-- logs/
```

B.3 Key Commands

B.3.1 FiftyOne Dataset Operations

```
import fiftyone as fo
import fiftyone.brain as fob

# Load dataset
dataset = fo.load_dataset("santa_ines_dron")

# Access embeddings
results = dataset.load_brain_results("clip_viz")
points = results.points  # Shape: (97, 2)

# Similarity search
view = dataset.sort_by_similarity(
    sample_id, k=10, brain_key="clip_similarity"
)

# Filter by tags
high_alt = dataset.match_tags("high-altitude")
closeups = dataset.match_tags("close-up")
```

B.3.2 Compilation

```
# Compile LaTeX document
cd research/paper
pdflatex main.tex
pdflatex main.tex  # Run twice for TOC

# Or use latexmk for automatic compilation
latexmk -pdf main.tex
```

B.4 FiftyOne Brain Keys

Brain Key	Type	Description
clip_viz	visualization	CLIP embeddings + UMAP 2D projection
clip_similarity	similarity	Similarity index for retrieval

B.5 GPS Extraction from Drone Images

GPS coordinates were extracted from EXIF metadata embedded in DJI drone images:

```
from PIL import Image
from PIL.ExifTags import TAGS, GPSTAGS

def get_gps_coords(image_path):
    """Extract GPS coordinates from EXIF data"""
    image = Image.open(image_path)
    exif_data = image._getexif()

    gps_info = exif_data.get('GPSInfo', {})

    # Convert DMS to decimal degrees
    def to_degrees(dms):
        d, m, s = dms
        return d + m/60.0 + s/3600.0

    lat = to_degrees(gps_info['GPSLatitude'])
    if gps_info['GPSLatitudeRef'] == 'S':
        lat = -lat

    lon = to_degrees(gps_info['GPSLongitude'])
    if gps_info['GPSLongitudeRef'] == 'W':
        lon = -lon

    return lat, lon
```

B.6 Satellite Map Integration

Google Maps Static API was used to obtain satellite imagery background:

```
import requests

API_KEY = "YOUR_API_KEY"
center_lat, center_lon = -34.4746, -70.9516
zoom = 18  # High zoom for agricultural detail

url = (f"https://maps.googleapis.com/maps/api/staticmap"
       f"?center={center_lat},{center_lon}"
       f"&zoom={zoom}&size=640x640"
       f"&maptype=satellite&key={API_KEY}")

response = requests.get(url)
satellite_img = plt.imread(BytesIO(response.content))
```

B.6.1 Coordinate Projection

GPS coordinates were converted to image pixel coordinates using Web Mercator projection:

```python
import numpy as np

def gps_to_pixels(lat, lon, center_lat, center_lon,
                  zoom, img_size):
    """Convert GPS to image pixel coordinates"""
    def lat_lon_to_world(lat, lon, zoom):
        lat_rad = np.radians(lat)
        n = 2 ** zoom
        x = (lon + 180) / 360 * n * 256
        y = (1 - np.log(np.tan(lat_rad) +
             1/np.cos(lat_rad)) / np.pi) / 2 * n * 256
        return x, y

    cx, cy = lat_lon_to_world(center_lat, center_lon, zoom)
    px, py = lat_lon_to_world(lat, lon, zoom)

    x = (px - cx) + img_size / 2
    y = (py - cy) + img_size / 2
    return x, y
```

B.7 Analysis Scripts Reference

Table 16: Python scripts for analysis pipeline

Script	Purpose
gps_satellite_map.py	Extract GPS from EXIF metadata
satellite_map.py	Generate satellite map with GPS overlay
cluster_samples.py	Select representative images per cluster
combined_figure.py	Create 4-panel combined analysis figure

Table 17: R scripts for statistical analysis

Script	Purpose
analysis.R	Embedding visualization, cluster statistics
similarity_analysis.R	Network analysis, cohesion metrics